Experiment 1.7

Mapped Course Outcomes-CO4

CO4: Deploy the concept to handle traffic and control on it

AIM:- Implement Router as DHCP server that can serve multiple VLAN's.

What will you learn:

How DHCP is used to assign IP addresses dynamically in the network .DHCP network creation with VLAN Protocol.

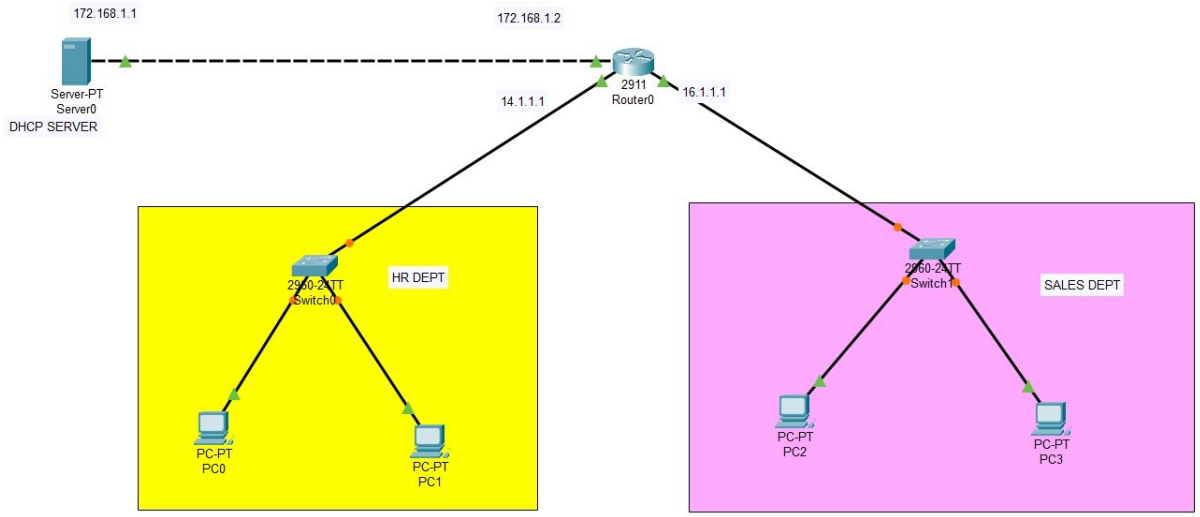
Prerequisites:-

1. An access to CISCO PACKET TRACER .

Dynamic Host Configuration Protocol (DHCP) is an application layer protocol used to distribute various network configuration parameters to devices on a TCP/IP network. – IP addresses, subnet masks, default gateways, DNS servers, etc. DHCP employs a client-server architecture; a DHCP client is configured to request network parameters from a DHCP server on the network. A DHCP server is configured with a pool of available IP addresses and assigns one of them to the DHCP client.[1]

A Cisco router can be configured as a DHCP server.

Topology:



Step by Step Procedure:

Here are the steps:

1. Exclude IP addresses from being assigned by DHCP by using the *ip dhcp excluded-address FIRST_IP LAST_IP*

2. Create a new DHCP pool with the *ip dhcp pool NAME* command.

3. Define a subnet that will be used to assign IP addresses to hosts with the *network SUBNET SUBNET_MASK* command.

4. Define the default gateway with the *default-router IP* command.

5. Define the DNS server with the *dns-server IP* address command.

6. (Optional) Define the DNS domain name by using the *ip domain-name NAME* command.

7. (Optional) Define the lease duration by using the *lease DAYS HOURS MINUTES* command.

If you don't specify this argument, the default lease time of 24 hours will be used.

Example Configuration.

Floor1(config)#ip dhcp excluded-address 192.168.0.1 192.168.0.50

Floor1(config)#ip dhcp pool Floor1DHCP

Floor1(dhcp-config)#network 192.168.0.0 255.255.255.0

Floor1(dhcp-config)#default-router 192.168.0.1

Floor1(dhcp-config)#dns-server 192.168.0.1

In the example above you can see that I've configured the DHCP server with the following parameters:

· the IP addresses from the **192.168.0.1 – 192.168.0.50** range will not be assigned to hosts.

· the DHCP pool was created and named **Floor1DHCP**

· the IP addresses assigned to the hosts will be from the **192.168.0.0/24** range

· the default gateway's IP address is **192.168.0.1**

· the DNS server's IP address is **192.168.0.1**

Conclusion & Discussion.

In the output above you can see that there is a single DHCP client that was assigned the IP address of **192.168.0.51**. Since we've excluded the IP addresses from the **192.168.0.1 – 192.168.0.50** range, the device got the first address available – **192.168.0.51**. To display information about the configured DHCP pools, you can use the *show ip dhcp pool* command

Further Reading:-

1. https://www.geeksforgeeks.org/dynamic-host-configuration-protocol-dhcp/